

Stickpoint — 8 Study Methods, Evidence, and App Implementation

THE 8 STUDY METHODS

1. Active Recall

Psychological evidence Active recall is one of the most extensively researched study methods in cognitive psychology. Roediger and Karpicke's landmark 2006 study published in Psychological Science found that students who tested themselves on material retained 50% more one week later than students who simply reread their notes. This is called the testing effect — the act of retrieving information from memory actually strengthens the memory itself. The harder your brain works to pull something out, the stronger that neural pathway becomes. Rereading by comparison creates a false sense of familiarity without building real retrieval strength.

How it helps students study better and more efficiently Instead of spending an hour rereading the same page, a student spends 20 minutes being tested on it and retains significantly more. It also immediately shows them what they don't know, so they stop wasting time on what they already have and focus on the actual gaps. For students preparing for AP exams or SATs where recall under pressure is everything, this is the most directly applicable method.

Who it suits based on the quiz Students who say they understand material while studying but go blank on the test. Students who currently reread notes or make flashcards. Students studying math, science, or history where factual retrieval is required. Students who say they learn best by testing themselves.

Step-by-step directions for real life

1. Read through your notes once, normally.
2. Close or flip over your notes so you cannot see them.
3. On a blank piece of paper or in a blank document, write out everything you can remember from what you just read.
4. Open your notes and compare. Circle or highlight everything you missed or got wrong.
5. Focus your next study pass only on what you missed.
6. Close your notes again and test yourself a second time.
7. Repeat until you can retrieve everything without looking.

Step-by-step how it works in the app

1. Student pastes their notes into the app.
 2. The app reads the notes and generates 8 to 10 question and answer flashcards from the most important concepts.
 3. Student sees the question side of the card only.
 4. Student thinks of or types their answer before tapping to reveal.
 5. Card flips to show the correct answer.
 6. Student rates themselves: easy, medium, or hard.
 7. Cards rated hard come back again later in the same session.
 8. At the end of the session a score shows how many they got right and which cards to focus on next time.
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2. Blurting

Psychological evidence Blurting is a direct application of the generation effect, first documented by Slamecka and Graf in 1978. Research consistently shows that information you generate yourself is remembered significantly better than information you simply read. When you force yourself to write out everything you know without looking, your brain must work harder to construct the memory, and that effort is what makes it stick. It also acts as a retrieval practice exercise, which shares the same evidence base as active recall.

How it helps students study better and more efficiently Blurting is fast. A student can blurt everything they know about a topic in under two minutes. It immediately reveals exactly where the gaps are without needing a teacher or a quiz. It is also a zero-setup method since all you need is a blank page, which makes it one of the most accessible methods for any student in any situation.

Who it suits based on the quiz Students who say they do not know what to focus on or where to start. Students who feel overwhelmed by how much there is to study. Students studying vocabulary-heavy subjects like languages or science. Students who say they understand material in class but it disappears by test day. Students who prefer writing over speaking.

Step-by-step directions for real life

1. Read through your notes or a section of your textbook.
2. Close everything completely.
3. On a blank piece of paper, write down every single thing you can remember about the topic. Do not stop, do not look, do not edit. Just write.
4. When you run out of things to write, open your notes.

5. Underline or circle everything in your notes that you did not write down.
6. Read only the things you missed.
7. Close your notes again and blurt a second time, trying to add what you missed.
8. Repeat until your blurt matches your notes.

Step-by-step how it works in the app

1. Student pastes their notes into the app.
 2. App shows the topic or subject title only, not the notes.
 3. A blank text box appears with the prompt: write down everything you know about this topic without looking.
 4. Student types freely for as long as they want.
 5. When they press done, the app compares their blurt to the original notes using AI.
 6. App highlights the key concepts they covered in green and the ones they missed in red.
 7. Student can review what they missed and blurt again.
 8. Each round their score improves showing them visible progress.
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3. Practice Testing

Psychological evidence Practice testing has the strongest evidence base of any study method. Dunlosky et al.'s 2013 landmark review in *Psychological Science in the Public Interest* rated practice testing as high utility, the highest rating given, and rated rereading as low utility, the lowest. Taking practice tests under realistic conditions, meaning timed and without notes, is the closest simulation to what a real exam demands. Research also shows that students who practice test consistently significantly outperform students who study more but test themselves less, even when total study time is equal.

How it helps students study better and more efficiently Practice testing shows students exactly what a real test will feel like, which reduces test anxiety because the actual test no longer feels unfamiliar. It also forces retrieval under pressure, which is the exact skill being tested. Students who only study passively are essentially training for a different game than the one they are being graded on.

Who it suits based on the quiz Students who say they do not know if their studying is working. Students who experience test anxiety. Students who need a score or measurable progress to stay motivated. Students studying any subject since practice testing works universally across all content types. Students preparing for standardized tests like the SAT or AP exams.

Step-by-step directions for real life

1. Find or create practice questions for your topic. Use past tests, textbook review questions, or generate your own.
2. Set a timer for a realistic amount of time, the same as the real test if possible.
3. Put your notes away completely and attempt every question without looking anything up.
4. When the timer ends, stop even if you are not finished.
5. Grade your own work against your notes or an answer key.
6. For every question you got wrong, write out the correct answer in your own words.
7. Do another round of practice testing on only the questions you got wrong.

Step-by-step how it works in the app

1. Student pastes their notes into the app.
 2. At the end of each study session, the app generates a five question mini quiz from the material covered.
 3. Quiz is timed, one to two minutes per question.
 4. Student answers each question without being able to see their notes.
 5. When submitted, the app grades the quiz and gives a score out of five.
 6. Correct answers are shown in green. Wrong answers show the correct answer alongside the student's answer.
 7. Score is saved so the student can track improvement across sessions.
 8. Any question answered wrong is flagged to come back in the next session.
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4. Feynman Technique

Psychological evidence The Feynman Technique is rooted in elaborative encoding, which is one of the most well-established principles in cognitive psychology. Research by Craik and Lockhart in 1972 showed that information processed at a deep semantic level, meaning you engage with meaning rather than surface features, is retained far longer than information processed shallowly. Explaining something in your own simple words forces deep processing. The generation effect also applies here since you are constructing the explanation yourself. Additionally research on peer teaching consistently shows that the act of teaching something consolidates the teacher's own understanding.

How it helps students study better and more efficiently The Feynman Technique immediately exposes the difference between recognizing something and actually understanding it. A student can read the definition of natural selection five times and feel like they know it. The moment they try to explain it simply without looking, they discover

exactly where the understanding breaks down. This saves time by targeting real gaps rather than giving students a false sense of preparation.

Who it suits based on the quiz Students who say they understand material in class but cannot explain it in their own words. Students studying humanities, biology, economics, or any conceptual subject. Students who say they learn best when things make sense rather than when they are memorized. Students who struggle with essay questions or free-response sections on tests.

Step-by-step directions for real life

1. Choose one concept from your notes, just one.
2. Put your notes away.
3. On a blank page write or say out loud: explain this concept as if you are teaching it to a 10-year-old who has never heard of it.
4. Write or speak freely until you run out of things to say.
5. Look back at your notes and find where your explanation was vague, wrong, or incomplete.
6. Go back and study only those gaps.
7. Try the explanation again from scratch.
8. You understand the concept when you can explain it simply without any gaps or vague language.

Step-by-step how it works in the app

1. Student pastes their notes into the app.
 2. The app identifies the key concepts in the notes and picks one to start with.
 3. A prompt appears: explain this concept in your own words as if you are teaching it to someone who has never heard of it.
 4. Student types their explanation freely in a text box.
 5. The AI reads the explanation and compares it to the original notes.
 6. App responds with specific feedback: what the student got right, what was missing, and what was unclear.
 7. Student can try again with the feedback in mind.
 8. Once the explanation is solid the app moves to the next concept.
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5. Concrete Examples

Psychological evidence The concreteness effect is a well-documented phenomenon in cognitive psychology. Research by Paivio's dual coding theory, developed in the 1970s and

extensively validated since, shows that information paired with a concrete image or real-world example is encoded in two ways simultaneously: verbally and visually. This double encoding significantly improves recall. Abstract information with no concrete anchor is much harder for the brain to retain because it has no hook to attach to. Research also shows that understanding a principle through an example transfers better to new problems than memorizing the principle alone.

How it helps students study better and more efficiently Rather than trying to memorize an abstract definition, a student who understands a concrete example of that concept can reconstruct the definition themselves, and also recognize the concept when it appears in a different form on a test. This is especially powerful in math and science where formulas and theories need to be applied, not just recalled.

Who it suits based on the quiz Students who say abstract concepts confuse them. Students studying math, science, economics, or philosophy where abstract principles are common. Students who say they connect to real life examples better than textbook definitions. Students who feel frustrated because they can read something multiple times and still not understand it.

Step-by-step directions for real life

1. After reading a concept or definition, stop and ask yourself: what is a real-world example of this?
2. The example should be something you have actually seen, heard, or experienced, not another textbook example.
3. Write the concept, then below it write your example.
4. Ask yourself: how does my example show the concept? Be specific.
5. If you cannot find an example, you do not understand the concept yet. Go back and reread.
6. Test yourself: read only the example and try to state the concept from it without looking.

Step-by-step how it works in the app

1. Student pastes their notes into the app.
2. After each concept card the app generates a real-world example or analogy that illustrates the concept.
3. The example is shown after the student has attempted to recall the concept themselves.
4. Student reads the example and the app asks: how does this example show the concept?
5. Student types a brief answer.

6. App confirms whether their connection is correct or clarifies where it missed.
 7. For math and science the app generates a sample problem as the concrete example and walks through why it applies.
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6. Pomodoro Timer

Psychological evidence The Pomodoro Technique is grounded in research on cognitive fatigue and attention. Attention research by Ariga and Lleras published in 2011 in *Cognition* found that brief mental breaks significantly improve sustained attention over longer periods. The brain is not designed for sustained deep focus for hours at a time. Research on the vigilance decrement shows that focused attention begins to degrade after 20 to 40 minutes without a break. Structured rest periods reset this and allow the next work session to begin at full cognitive capacity rather than diminished capacity.

How it helps students study better and more efficiently A student who studies for two hours while gradually losing focus retains far less than a student who does four focused 25-minute sessions with breaks. The Pomodoro method also removes the psychological resistance to starting because the endpoint is always visible. It is easier to commit to 25 minutes than to commit to studying until you are done.

Who it suits based on the quiz Students who say they get distracted and lose focus quickly. Students who say they check their phone while studying. Students who do not have a real study pattern. Students who feel overwhelmed by how much material there is. Students who struggle to start studying at all.

Step-by-step directions for real life

1. Put your phone on do not disturb and place it face down or in another room.
2. Open only the tab or material you need. Nothing else.
3. Set a timer for 25 minutes.
4. Study with complete focus until the timer goes off. If a distraction pops into your head, write it down and ignore it until the break.
5. When the timer goes off, stop immediately even if you are mid-sentence.
6. Take a real five minute break. Stand up, walk around, drink water. Do not scroll your phone.
7. After the break, start a new 25-minute round.
8. After four rounds take a longer 20-minute break.

Step-by-step how it works in the app

1. When the student starts a study session the Pomodoro timer activates automatically in the background.

2. A small progress bar at the top of the screen shows how much time is left in the current session.
 3. The student scrolls through their study cards normally while the timer runs.
 4. At 25 minutes the app pauses automatically and shows a break screen with a fun fact or subject-related meme.
 5. A five minute countdown runs on the break screen.
 6. When the break ends the app resumes the session automatically.
 7. After four full sessions the app shows a longer break screen and a session summary.
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7. Confidence Rating

Psychological evidence Confidence rating is a metacognitive study tool. Metacognition, which is thinking about your own thinking, is one of the most well-researched areas in educational psychology. Flavell's foundational work in 1979 established that students who accurately monitor their own understanding learn significantly more effectively than students who do not. Research by Dunning and Kruger also shows that students who lack metacognitive skills tend to dramatically overestimate how well they know material, which leads them to stop studying too early. Confidence rating forces students to accurately assess their own knowledge after each piece of material, which leads to better allocation of study time.

How it helps students study better and more efficiently Confidence rating stops students from spending equal time on things they already know and things they do not. It immediately redirects attention to material that needs more work and gives the student real-time data about where they are weakest. It also reduces the Dunning-Kruger effect by making overconfidence visible in the data.

Who it suits based on the quiz Students who do not know what to focus on or feel overwhelmed by the amount of material. Students who run out of time studying and guess incorrectly about what to prioritize. Students who feel confident before a test and still do poorly. Students who want to feel in control of their own study session.

Step-by-step directions for real life

1. After testing yourself on any piece of material, immediately rate your confidence on a simple scale: easy meaning you knew it instantly, medium meaning you got it but were not sure, or hard meaning you did not know it or got it wrong.
2. Keep a list of all your hard and medium ratings.
3. Your next study session should begin with everything rated hard, then medium.
4. Everything rated easy gets reviewed briefly at the very end.
5. Re-rate after each review and adjust accordingly.

Step-by-step how it works in the app

1. After each card in the study session three buttons appear: Easy, Medium, Hard.
 2. Student taps one honestly.
 3. Cards rated Hard come back again before the current session ends.
 4. Cards rated Medium come back once at the end of the session.
 5. Cards rated Easy are set aside.
 6. At the end of the session the app shows a breakdown: how many easy, medium, and hard cards the student had.
 7. The next time the student opens the app the session starts with their previous hard cards before introducing new ones.
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8. Elaborative Interrogation

Psychological evidence Elaborative interrogation was studied extensively by Pressley, Symons, McDaniel, Snyder, and Turnure in 1988, and follow-up research has consistently validated it. The method works by activating prior knowledge and forcing the brain to connect new information to what it already knows. Research shows that asking why-questions during studying significantly improves retention compared to reading the same material without questioning. The mechanism is that answering why forces the brain to process information at a deeper semantic level and integrate it into existing knowledge structures rather than storing it in isolation.

How it helps students study better and more efficiently Students who ask why something is true understand the logic behind a fact, not just the fact itself. This means they can reconstruct the answer on a test even if they cannot remember it exactly, because they understand how it follows from something they do know. It also makes studying more engaging because it turns passive reading into active sense-making.

Who it suits based on the quiz Students who say they want to understand why something is true rather than just memorize it. Students studying history, biology, economics, or any subject with cause and effect relationships. Students who say they forget things that do not make sense to them. Students who enjoy thinking deeply about ideas. Students who are curious and find rote memorization boring.

Step-by-step directions for real life

1. Read a fact or concept from your notes.
2. Immediately ask yourself: why is this true? Do not move on until you have an answer.
3. If you do not know why, look it up. Then ask why again.
4. Also ask: how does this connect to what I already know about this topic?

5. Write your why-answer below the fact in your notes.
6. Later when you review, try to reconstruct the fact from the why. If you remember the reason, you can usually reconstruct the fact even without memorizing it directly.

Step-by-step how it works in the app

1. Student pastes their notes into the app.
 2. After a concept or fact card, the app generates a why-question: why do you think this is true, or how does this connect to what came before?
 3. Student types their answer in a text box.
 4. The AI evaluates the answer and responds with whether the reasoning is correct, what was missing, and any additional connections the student may have missed.
 5. Student can read the full explanation if they are stuck.
 6. The question and the student's answer are saved so they can review their reasoning later.
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SUBJECT TO METHOD MAPPING

Math and Science Primary: Active Recall, Practice Testing, Concrete Examples Secondary: Elaborative Interrogation, Confidence Rating Reason: These subjects require accurate retrieval of formulas and procedures, application to new problems, and deep understanding of why methods work.

History and Humanities Primary: Feynman Technique, Elaborative Interrogation, Blurting Secondary: Active Recall, Confidence Rating Reason: These subjects reward understanding connections and causes, being able to explain concepts in your own words, and recalling large amounts of narrative information.

Language Primary: Active Recall, Blurting, Confidence Rating Secondary: Practice Testing Reason: Language learning is heavily vocabulary-based which requires fast, accurate retrieval of words and their meanings, with confidence rating helping students know which words need more repetition.

English and Writing Primary: Feynman Technique, Elaborative Interrogation, Concrete Examples Secondary: Blurting Reason: English rewards deep understanding of texts and ideas, the ability to articulate arguments clearly, and connecting abstract themes to concrete evidence.

Multiple Subjects Primary: Pomodoro Timer, Confidence Rating, Active Recall Secondary: Blurting Reason: Students with multiple subjects need structure and time management above everything else, with confidence rating helping them prioritize across subjects.

REVISED QUIZ QUESTIONS

These questions are designed to map student answers directly to the 8 methods above.

Q1 — What subject are you focusing on right now?

- Math or Science
- History or Humanities
- A language like Spanish or French
- English or Writing
- Multiple subjects at once

This is asked first because subject immediately narrows which methods the app should weight most heavily in the recommendation.

Q2 — What does studying usually look like for you right now?

- I reread my notes or textbook until I feel ready
- I write out everything I remember on a blank page
- I make flashcards or questions and test myself
- I ask AI or watch videos to explain things to me
- I do practice problems or past test questions

Rereading → needs active recall, blurting, or practice testing. Writing from scratch → already blurting, build on it. Flashcards → active recall with confidence rating. AI or videos → Feynman and concrete examples. Practice problems → practice testing and elaborative interrogation.

Q3 — What is your biggest problem when you study?

- I understand it while I study but forget it on the actual test
- I always run out of time and end up cramming
- I get distracted and lose focus after a few minutes
- I don't know what to focus on or where to even start
- I can read something five times and still not really get it

Forget on test → active recall, practice testing. Cramming → Pomodoro to use time well. Distracted → Pomodoro. Don't know where to start → confidence rating, blurting. Don't understand → Feynman, elaborative interrogation, concrete examples.

Q4 — When you actually remember something weeks later, what usually helped?

- I explained it out loud to myself or someone else
- I tested myself on it over and over
- I connected it to a real-life example that made it click
- I kept asking myself why it works that way
- I wrote out everything I knew from memory without looking

Explained out loud → Feynman. Tested repeatedly → active recall, practice testing. Real-life example → concrete examples. Kept asking why → elaborative interrogation. Wrote from memory → blurting.

Q5 — Which of these sounds most like how you feel when you study?

- I lose focus and drift off after about ten minutes
- I feel okay but I'm never sure if the studying is actually working
- I get frustrated when I don't understand something and give up
- I feel overwhelmed by how much there is and don't know what to prioritize
- I feel confident while studying but nervous when the test actually comes

Lose focus → Pomodoro. Not sure if it's working → practice testing, confidence rating. Get frustrated → concrete examples, Feynman. Overwhelmed → confidence rating, blurting. Confident while studying, nervous on test → active recall, practice testing.

Q6 — How far in advance do you usually start studying for a big test?

- Night before or even the day of
- Two to three days before
- About a week or more before
- I don't really have a consistent pattern

Night before → Pomodoro to make the most of limited time, blurting for speed. Two to three days → active recall, practice testing. A week out → full session with all methods. No pattern → confidence rating and Pomodoro to build structure.

Q7 — What would make studying feel like it's actually working for you?

- Seeing a score go up or a measurable improvement
- Finally understanding something that confused me
- Remembering things without having to look them up
- Staying focused for the whole session without getting distracted
- Knowing exactly what I need to work on and what I can stop worrying about

Score going up → practice testing, confidence rating. Finally understanding → Feynman, concrete examples, elaborative interrogation. Remembering without looking → active recall, blurting. Staying focused → Pomodoro. Knowing what to work on → confidence rating, blurting.

HOW THE SCORING WORKS

Each answer adds points to one or more of the 8 methods. At the end of all 7 questions the app ranks all 8 methods by total score and recommends the top two or three. It never recommends just one because most students benefit from a combination. The subject from Q1 also applies a multiplier that boosts subject-appropriate methods. For example a student studying history who cannot explain things in their own words and gets frustrated would receive Feynman as their primary method, elaborative interrogation as their secondary, and blurting as a quick daily habit. The app then explains why each was recommended using specific language that reflects what the student actually answered, not generic descriptions.